

Geometry (7 th std Term 2)

Choose the appropriate answer

- Triangle is formed by joining three _____ points.
(a) Collinear (b) Non – collinear (c) Equal (d) None of these
Ans: (b) Non – collinear
- A triangle has _____ vertices and _____ sides.
(a) 3, 2 (b) 2, 3 (c) 3, 3 (d) None of these
Ans: (c) 3, 3
- A point where two sides of a triangle meet is known as _____ of a triangle.
(a) Vertex (b) Angle (c) Corner (d) Intersection
Ans: (a) Vertex
- Each angle of an equilateral triangle is of _____ measure.
(a) 90° (b) 60° (c) 45° (d) 30°
Ans: (b) 60°
- A triangle has angle measurements of 29° , 65° and 86° . Then it is _____ triangle.
(a) an acute angled (b) a right angled (c) an obtuse angled (d) a scalene
Ans: (a) an acute angled
- A triangle has angle measurements of 30° , 30° and 120° . Then it is _____ triangle.
(a) an acute angled (b) scalene (c) obtuse angled (d) right angled
Ans: (c) obtuse angled
- Which of the following can be the sides of a triangle?
(a) 5,9,14 (b) 7,7,15 (c) 1, 2, 4 (d) 3, 6, 8
Ans: (a) 5,9,14

The sum of any two sides must be greater than the third side
 $(5+9 > 14)$ is false, so this is not correct

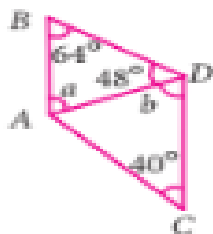
- Ezhil wants to fence his triangular garden. If two of the sides measure 8 feet and 14 feet then the length of the third side is _____.
(a) 11ft (b) 6ft (c) 5ft (d) 22ft
Ans: (c) 5ft
- The angles of a triangle are in the ratio 2:3:4. Then the angles are
(a) 20, 30, 40 (b) 40, 60, 80 (c) 80, 20, 80 (d) 10, 15, 20
Ans: (b) 40, 60, 80

10. One of the angles of a triangle is 65° . If the difference of the other two angles is 45° , then the two angles are

- (a) $85^\circ, 40^\circ$ (b) $70^\circ, 25^\circ$ (c) $80^\circ, 35^\circ$ (d) $80^\circ, 135^\circ$

Ans: (c) $80^\circ, 35^\circ$

11. In the given figure, AB is parallel to CD. Then the value of b is



- (a) 112° (b) 68° (c) 102° (d) 62°

Ans: (b) 68°

12. In the given figure, which of the following statement is true?

- (a) $x + y + z = 180^\circ$ (b) $x + y + z = a + b + c$
 (c) $x + y + z = 2(a + b + c)$ (d) $x + y + z = 3(a + b + c)$

Ans: (c) $x + y + z = 2(a + b + c)$

13. An exterior angle of a triangle is 70° and two interior opposite angles are equal. Then measure of each of these angle will be

- (a) 110° (b) 120° (c) 35° (d) 60°

Ans: (c) 35°

14. In a $\triangle ABC$, $AB = AC$. The value of x is _____.

- (a) 80° (b) 100° (c) 130° (d) 120°

Ans: (c) 130°

15. If an exterior angle of a triangle is 115° and one of the interior opposite angles is 35° , then the other two angles of the triangle are

- (a) $45^\circ, 60^\circ$ (b) $65^\circ, 80^\circ$ (c) $65^\circ, 70^\circ$ (d) $115^\circ, 60^\circ$

Ans: (b) $65^\circ, 80^\circ$

16. If two plane figures are congruent then they have

- (a) Same size (b) Same shape (c) Same angle (d) Same shape and same size

Ans: (d) Same shape and same size

17. Which of the following methods are used to check the congruence of plane figures?

- (a) Translation method (b) Superposition method
 (c) Substitution method (d) Transposition method

Ans: (b) Superposition method

18. Which of the following rule is not sufficient to verify the congruency of two triangles.

- (a) SSS rule (b) SAS rule (c) SSA rule (d) ASA rule

Ans: (c) SSA rule

19. Two students drew a line segment each. What is the condition for them to be congruent?
- They should be drawn with a scale.
 - They should be drawn on the same sheet of paper.
 - They should have different lengths.
 - They should have the same length.

Ans: (d) They should have the same length

20. In the given figure, $AD=CD$ and $AB=CB$. Identify the other three pairs that are equal.

- $\angle ADB = \angle CDB$, $\angle ABD = \angle CBD$, $BD = BD$
- $AD = AB$, $DC = CB$, $BD = BD$
- $AB = CD$, $AD = BC$, $BD = BD$
- $\angle ADB = \angle CDB$, $\angle ABD = \angle CBD$, $\angle DAB = \angle DBC$

Ans: (a) $\angle ADB = \angle CDB$, $\angle ABD = \angle CBD$, $BD = BD$

21. In $\triangle ABC$ and $\triangle PQR$, $\angle A=50^\circ = \angle P$, $PQ=AB$, and $PR=AC$. By which property $\triangle ABC$ and $\triangle PQR$ are congruent?

- SSS property
- SAS property
- ASA property
- RHS property

Ans: (b) SAS property

22. Two triangles are congruent if 'The three sides of the one are equal to the three corresponding sides of the other.' The above statement is known as

- SSS congruence of two triangles
- SAS congruence of two triangles
- ASA congruence of two triangles
- RHS congruence of two right-angled triangles

Ans: (a) SSS congruence of two triangles

23. Two triangles are congruent if two angles and the side included between them in one triangle is equal to the two angles and the side included between them of the other triangle. This is known

- RHS congruence criterion
- ASA congruence criterion
- SAS congruence criterion
- SSS congruence criterion

Ans: (b) ASA congruence criterion

24. The symbol for congruence is

- $\equiv$
- $\cong$
- $\leftrightarrow$
- $=$

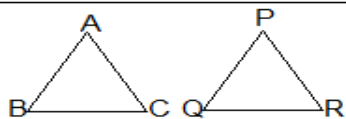
Ans: (b) $\cong$

25. If $\triangle ABC \cong \triangle PQR$, then which of the following is not true?

- $BC=PQ$
- $AC=PR$
- $QR=BC$
- $AB=PQ$

Ans: (a) $BC=PQ$

$ABC \cong POR$ Thus corresponding sides are equal.
Hence, $AB=PQ$ $AC=PR$ $BC=QR$
Hence, $BC=PQ$ is not true for the triangles.



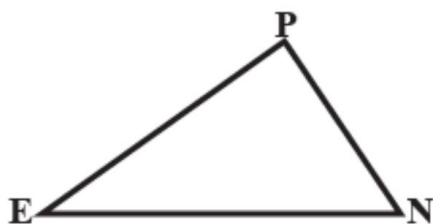
26. If $\triangle ABC = \triangle PQR$, then $\angle B$ corresponds to

- $\angle P$
- $\angle Q$
- $\angle R$
- None of these

Ans: (b) $\angle Q$

$\triangle ABC = \triangle PQR$, Then $\angle A = \angle P$ $\angle B = \angle Q$ $\angle C = \angle R$

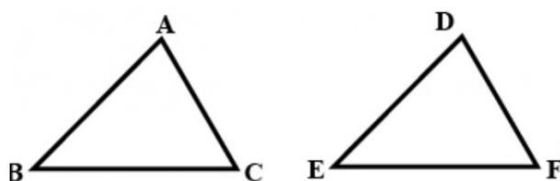
27. You want to show that $\triangle ART \cong \triangle PEN$ If it is given that $\angle T = \angle N$ and you are to use SAS criterion, you need to have $RT =$



- (a) EN (b) AT (c) EP (d) None of these

Ans: (a) EN

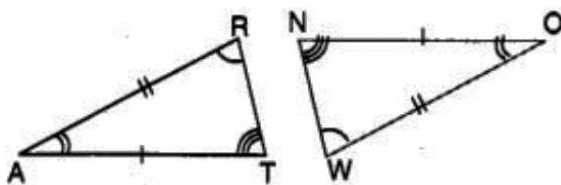
28. Which congruence criterion do you use in the following? Given $AC = DF$ $AB = DE$ $BC = EF$ So, $\triangle ABC \cong \triangle DEF$



- (a) SSS (b) SAS (c) ASA (d) RHS

Ans: (a) SSS

29. In the following figure, the two triangles are congruent. The corresponding parts are marked. We can write $\triangle RAT = ?$

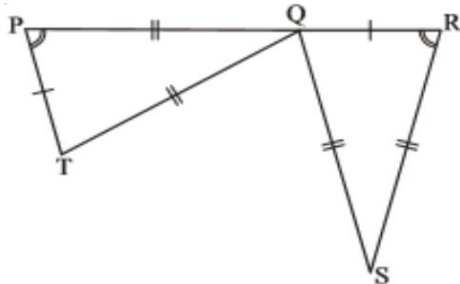


- (a) $\triangle WON$ (b) $\triangle WNO$ (c) $\triangle OWN$ (d) $\triangle ONW$

Ans: (a) $\triangle WON$

$A \leftrightarrow O, R \leftrightarrow W, T \leftrightarrow N$. We can write, $\triangle RAT \cong \triangle WON$ [By SAS congruence rule]

30. Complete the congruence statement $\triangle QRS$

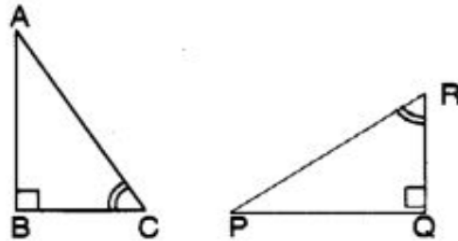


- (a) $\triangle TPQ$ (b) $\triangle TQP$ (c) $\triangle QTP$ (d) $\triangle QPT$

Ans: (a) $\triangle TPQ$

In the given figure, $QR=TP$ $QS=TQ$ $RS=PQ$ $\angle P=\angle R$
 $\Rightarrow \triangle QRS \cong \triangle TPQ$ (by SSS congruence)

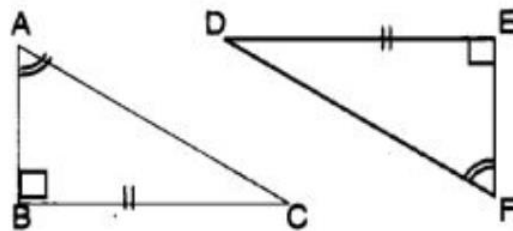
31. If $\triangle ABC$ and $\triangle PQR$ are to be congruent, name one additional pair of corresponding parts



- (a) $BC = PR$ (b) $BC = PQ$ (c) $BC = QR$ (d) None of these
Ans: (c) $BC = QR$

The additional corresponding part is $BC = RQ$ by ASA congruence rules.

32. By which congruence, is $\triangle ABC = \triangle FED$?



- (a) SSS (b) SAS (c) ASA (d) RHS
Ans: (c) ASA

$\angle B = \angle E$, $BC = ED$, $\angle D = \angle C$ (By angle sum property of a triangle).

33. Number of elements of a triangle is:

- (a) 2 (b) 3 (c) 4 (d) 6
Ans: (d) 6

A triangle has total six elements. They are its three angles and three sides.

34. Two figures are said to be congruent if they have exactly the same:

- (a) Length and Width (b) Shape and size (c) Area (d) Perimeter
Ans: (b) Shape and size

Two plane figures are congruent to each other, if trace copy of one of the figures covers the other figure completely.

35. One of the angles of a triangle is 50° and the other two angles are equal. Find the measure of each of the equal angles.

- (a) $65^\circ, 65^\circ$ (b) $50^\circ, 50^\circ$ (c) $45^\circ, 45^\circ$ (d) $65^\circ, 50^\circ$
Ans: (a) $65^\circ, 65^\circ$

$$x + x + 50^\circ = 180^\circ, 2x = 130^\circ, x = 65^\circ$$

36. $\triangle ABC$ is right triangle in which $\angle A = 90^\circ$ and $AB = AC$. The values of $\angle B$ and $\angle C$ will be
 (a) $\angle B = \angle C = 30^\circ$ (b) $\angle B = \angle C = 50^\circ$ (c) $\angle B = \angle C = 45^\circ$ (d) $\angle B = \angle C = 60^\circ$
Ans: (c) $\angle B = \angle C = 45^\circ$

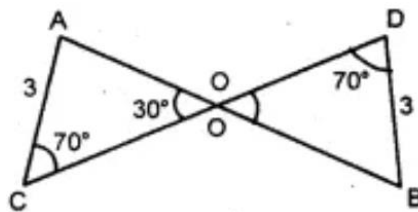
37. Two students drew a line segment each. What is the condition for them to be congruent?
 (a) They should be drawn with a scale
 (b) They should be drawn on the same sheet of paper
 (c) They should have different lengths
 (d) They should have the same length
Ans: (d) They should have the same length

38. A triangle in which all three sides are of equal lengths is called _____.
 (a) Isosceles (b) Equilateral (c) Scalene (d) None of these
Ans: (b) Equilateral

39. If the vertical angle of an isosceles triangle is 40° , then a measure of other two angles will be
 (a) $60^\circ, 60^\circ$ (b) $80^\circ, 80^\circ$ (c) $70^\circ, 70^\circ$ (d) $45^\circ, 45^\circ$
Ans: (c) $70^\circ, 70^\circ$

40. Which of the following examines the congruence of plane figures?
 (a) Trial and error method (b) Superposition method
 (c) Substitution method (d) Transposition method
Ans: (b) Superposition method

41. In the given figure, say congruency of two triangles.



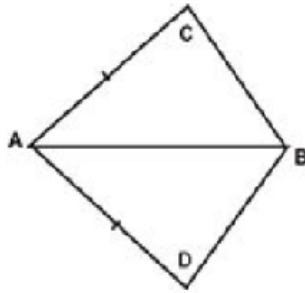
- (a) $\triangle AOC \cup \triangle BOD$ (b) $\triangle AOC \neq \triangle BOD$ (c) $\triangle AOC \cup \triangle OBD$ (d) None of these
Ans: (a) $\triangle AOC \cup \triangle BOD$

42. The angles of a triangle are in the ratio 2:3:5. What is the largest angle of the triangle?
 (a) 54° (b) 36° (c) 90° (d) 100°
Ans: (c) 90°

$$\text{Hence } \angle A + \angle B + \angle C = 180^\circ \Rightarrow 2x + 3x + 5x = 180^\circ \Rightarrow x = 18^\circ$$

$$\therefore \text{Angles are } 36^\circ, 54^\circ, 90^\circ. \text{ The largest angle of the triangle } 90^\circ$$

43. In the quadrilateral ABCD, $AC = AD$ and AB bisect $\angle A$ and $\triangle ABC \cong \triangle ABD$. The relation between BC and BD is



- (a) $BC < BD$ (b) $BC > BD$ (c) $BC = BD$ (d) None of these

Ans: (c) $BC = BD$

44. How many altitudes can a triangle have?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (c) 3

45. How many medians a triangle can have?

- (a) 2 (b) 1 (c) 3 (d) 0

Ans: (c) 3

46. A/an _____ connect a vertex of a triangle to the midpoint of the opposite side.

- (a) Altitude (b) Median (c) Vertex (d) None of these.

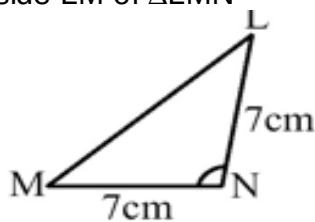
Ans: (b) Median

47. How many altitude can a triangle have?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (c) 3

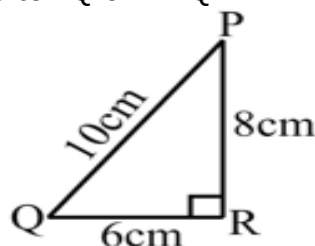
48. Angle opposite to the side LM of $\triangle LMN$



- (a) $\angle N$ (b) $\angle L$ (c) $\angle M$ (d) None of these

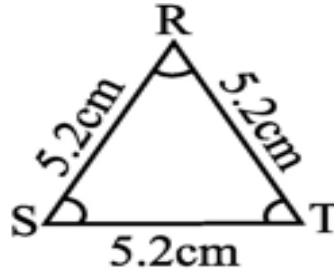
Ans: (a) $\angle N$

49. Side opposite to the vertex Q of $\triangle PQR$



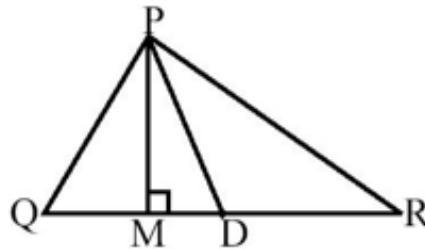
- (a) PQ (b) QR (c) PR (d) None of these
Ans: (c) PR

50. Vertex opposite to the side RT of $\triangle RST$



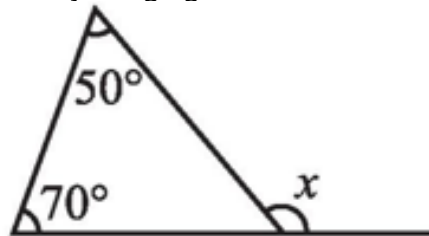
- (a) S (b) T (c) R (d) None of these
Ans: (a) S

51. In $\triangle PQR$, PM is



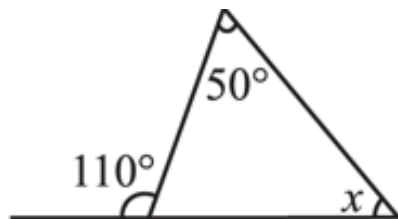
- (a) Median (b) Altitude (c) Bisector (d) Side
Ans: (b) Altitude

52. Find the value of x in the adjoining figure.



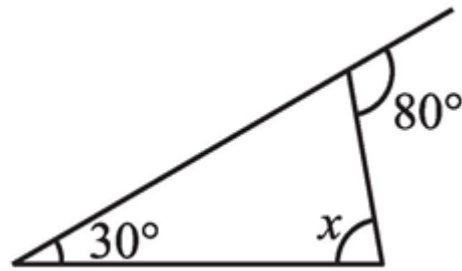
- (a) 50° (b) 70° (c) 120° (d) 180°
Ans: (c) 120°

53. Find the value of x



- (a) 60° (b) 110° (c) 50° (d) 180°
Ans: (a) 60°

54. Find the value of x



- (a) 50° (b) 60° (c) 90° (d) 80°

Ans: (a) 50°

55. A triangle in which two sides are of equal lengths is called ----- triangle.

- (a) Equilateral (b) Isosceles (c) Scalene (d) Acute angled

Ans: (b) Isosceles

56. The sum of the lengths of any two sides of a triangle is _____ the third side of the triangle.

- (a) Less than (b) Greater than (c) Double (d) Half

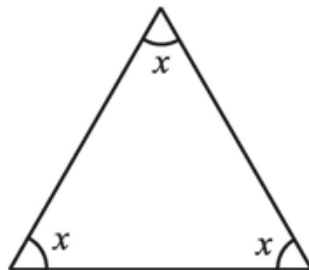
Ans: (b) Greater than

57. In the Pythagoras property, the triangle must be _____.

- (a) Acute angled (b) Right angled (c) Obtuse angled (d) None of these.

Ans: (b) Right angled

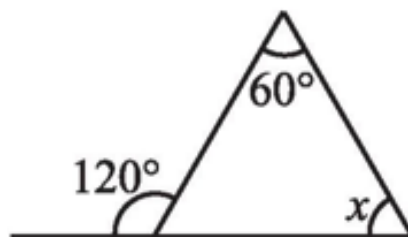
58. Find the value of x in this figure.



- (a) 40° (b) 60° (c) 35° (d) 180°

Ans: (b) 60°

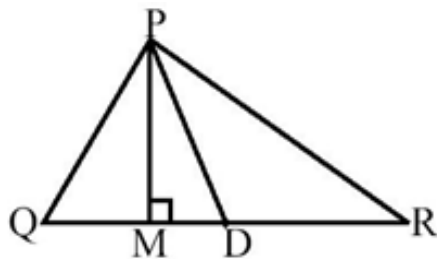
59. Find the value of x in given figure



- (a) 180° (b) 55° (c) 90° (d) 60°

Ans: (d) 60°

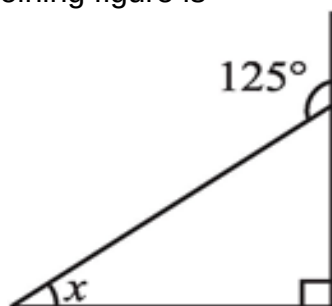
60. In $\triangle PQR$, PD is



- (a) Median (b) Altitude (c) Bisector (d) Side

Ans: (a) Median

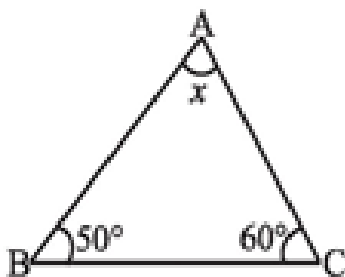
61. The value of x in the adjoining figure is



- (a) 125° (b) 90° (c) 180° (d) 35°

Ans: (d) 35°

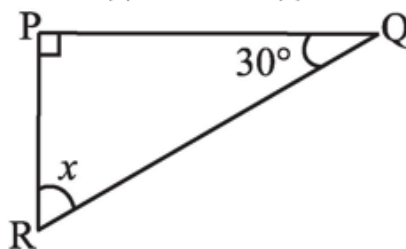
62. Find the value of unknown x in the adjoining figure.



- (a) 50° (b) 60° (c) 70° (d) 90°

Ans: (c) 70°

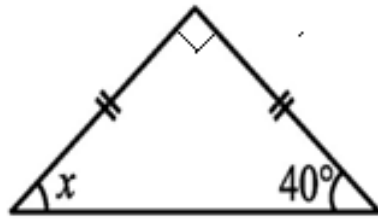
63. What is the value of x in the below figure.



- (a) 50° (b) 60° (c) 30° (d) 60°

Ans: (b) 60°

64. What is the measure of angle x



- (a) 90° (b) 50° (c) 180° (d) 120°

Ans: (b) 50°

65. $\triangle ABC$ is right-angled at C. If AC = 5 cm and BC = 12 cm find the length of AB.

- (a) 7 cm (b) 17 cm (c) 13 cm (d) None of these.

Ans: (c) 13 cm

According to Pythagorean theorem

$$AB^2 = 25 \text{ cm}^2 + 144 \text{ cm}^2 = \sqrt{169 \text{ cm}^2} \Rightarrow AB = 13 \text{ cm.}$$

66. PQR is a triangle right angled at P. If PQ = 3 cm and PR = 4 cm, find QR.

- (a) 7 cm (b) 1 cm (c) 5 cm (d) None of these.

Ans: (c) 5 cm

67. Which is the longest side in the triangle PQR right angled at P?

- (a) PQ (b) QR (c) PR (d) None of these.

Ans: (b) QR

68. Which is the longest side in the triangle ABC right angled at B?

- (a) AB (b) BC (c) AC (d) None of these.

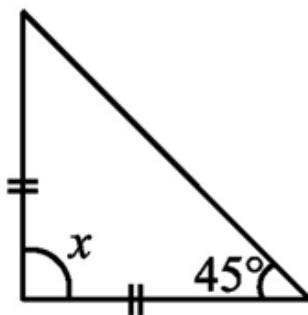
Ans: (c) AC

69. Which is the longest side of a right triangle?

- (a) Perpendicular (b) Base (c) Hypotenuse (d) None of these.

Ans: (c) Hypotenuse

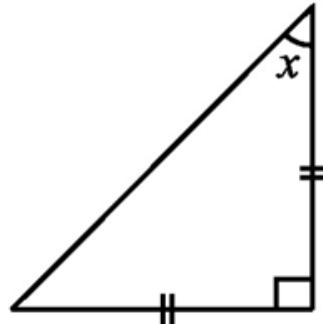
70. What is the measure of angle x



- (a) 90° (b) 60° (c) 180° (d) 120°

Ans: (a) 90°

71. The value of x in the adjoining figure is



- (a) 25° (b) 90° (c) 45° (d) 35°

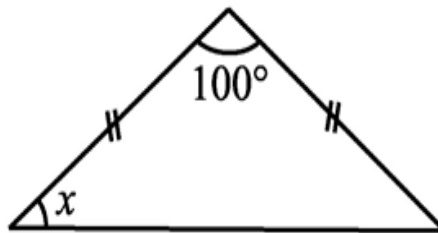
Ans: (c) 45°

72. A triangle in which all three sides are of equal lengths is called _____.

- (a) Equilateral (b) Isosceles (c) Scalene (d) Acute angled triangle

Ans: (a) Equilateral

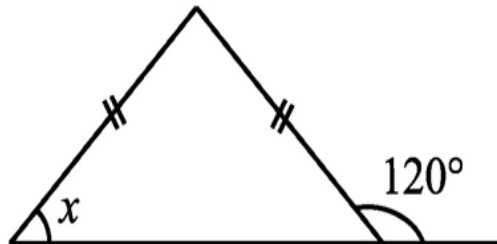
73. Find angle x in below figure:



- (a) 25° (b) 90° (c) 40° (d) 30°

Ans: (c) 40°

74. Find angle x in below figure:



- (a) 90° (b) 60° (c) 80° (d) 40°

Ans: (b) 60°

75. In a $\triangle ABC$, $\angle A = 35^\circ$ and $\angle B = 65^\circ$, then the measure of $\angle C$ is

- (a) 50° (b) 80° (c) 30° (d) 60°

Ans: (b) 80°

76. The hypotenuse of a right triangle is 17 cm long. If one of the remaining two sides is 8 cm in length, then the length of the other side is

- (a) 15 cm (b) 12 cm (c) 13 cm (d) None of these.

Ans: (a) 15 cm

$$8^2 + b^2 = 17^2 \Rightarrow 64 + b^2 = 289.$$

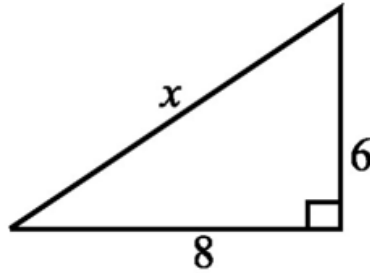
$$b^2 = 289 - 64 = 225 \Rightarrow b = \sqrt{225} = 15 \text{ cm.}$$

77. How many acute angles can a right triangle have?

- (a) 1 (b) 2 (c) 3 (d) 0

Ans: (b) 2

78. Find the unknown length x in the adjoining figure.



- (a) 11 (b) 9 (c) 10 (d) None of these

Ans: (c) 10

79. In a $\triangle PQR$, $\angle R = 105^\circ$ and $\angle Q = 40^\circ$, then the measure of $\angle P$ is

- (a) 45° (b) 80° (c) 35° (d) 60°

Ans: (c) 35°

80. In a $\triangle ABC$, $\angle A = 72^\circ$ and $\angle B = 63^\circ$, then the measure of $\angle C$ is

- (a) 45° (b) 80° (c) 30° (d) 60°

Ans: (a) 45°

81. In a $\triangle XYZ$, $\angle X = 90^\circ$ and $\angle Z = 48^\circ$, then the measure of $\angle Y$ is

- (a) 45° (b) 40° (c) 41° (d) 42°

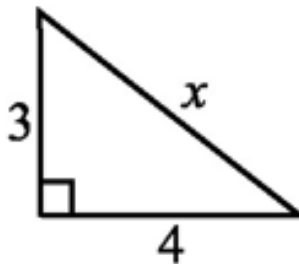
Ans: (d) 42°

82. One of the acute angle of a right triangle is 36° , then the other acute angle is

- (a) 55° (b) 54° (c) 51° (d) 52°

Ans: (b) 54°

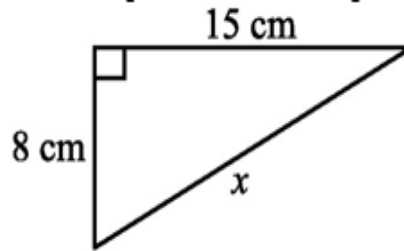
83. Find the unknown length x in the adjoining figure.



- (a) 5 (b) 7 (c) 3 (d) 4

Ans: (a) 5

84. Find the unknown length x in the below figure.



- (a) 15 (b) 17 (c) 13 (d) 14

Ans: (b) 17

85. The acute angles of right triangle are in the ratio 2: 1. Find the measure of each of these angles.

- (a) 55° and 35° (b) 60° and 30° (c) 50° and 40° (d) 45° and 45°

Ans: (b) 60° and 30°

86. One of the angles of a triangle is 100° and the other two angles are equal. Find the measure of each of these equal angles.

- (a) 45° (b) 40° (c) 41° (d) 42°

Ans: (b) 40°

87. Two angles are congruent if they have

- (a) Same name (b) Equal measures (c) Unequal measures (d) None of these

Ans: (b) Equal measures

88. Which of the following is not a congruence criterion?

- (a) SSS (b) SAS (c) ASA (d) None of these

Ans: (d) None of these

89. If a $\triangle ABC \cong \triangle PQR$, then AB is equal to

- (a) QR (b) PQ (c) PR (d) None of these

Ans: (b) PQ

90. In $\triangle ABC$ and $\triangle PQR$, $AB = 4\text{ cm}$, $BC = 5\text{ cm}$, $AC = 6\text{ cm}$ and $PQ = 4\text{ cm}$, $QR = 5\text{ cm}$, $PR = 6\text{ cm}$, then which of the following is true?

- (a) $\triangle ABC \cong \triangle QRP$ (b) $\triangle ABC \cong \triangle PQR$ (c) $\triangle ABC \cong \triangle PRQ$ (d) $\triangle ABC \cong \triangle QPR$

Ans: (b) $\triangle ABC \cong \triangle PQR$

91. If $\triangle DEF \cong \triangle BCA$, then the part of $\triangle BCA$ that correspond to $\angle E$ is

- (a) $\angle A$ (b) $\angle B$ (c) $\angle C$ (d) None of these

Ans: (c) $\angle C$

92. If $\triangle DEF \cong \triangle ACB$, then the part of $\triangle ACB$ that correspond to $\angle F$ is

- (a) $\angle A$ (b) $\angle B$ (c) $\angle C$ (d) None of these

Ans: (b) $\angle B$

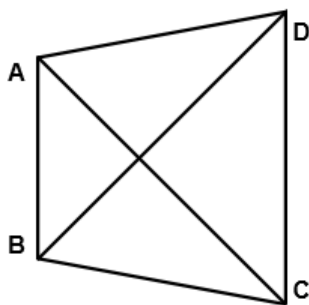
93. $\triangle ABC$ and $\triangle PQR$ are congruent under the correspondence: $ABC \leftrightarrow RQP$, then the part of $\triangle ABC$ that correspond to $\angle P$ is

- (a) $\angle A$ (b) $\angle B$ (c) $\angle C$ (d) None of these
Ans: (c) $\angle C$

94. $\triangle ABC$ and $\triangle PQR$ are congruent under the correspondence: $BCA \leftrightarrow RPQ$, then the part of $\triangle ABC$ that correspond to $\angle Q$ is

- (a) $\angle A$ (b) $\angle B$ (c) $\angle C$ (d) None of these
Ans: (b) $\angle B$

95. In the below quadrilateral ABCD, $AD = BC$ and $\angle DAB = \angle CBA$. If $\angle ABD \cong \angle BAC$. The relation between $\angle ABD$ and $\angle BAC$ is



- (a) $\angle ABD > \angle BAC$ (b) $\angle ABD < \angle BAC$ (c) $\angle ABD = \angle BAC$ (d) $\angle ABD = (1 / 2) \angle BAC$

Ans: (c) $\angle ABD = \angle BAC$

96. $\triangle ABC$ is right triangle in which $\angle A = 90^\circ$ and $AB = AC$. The values of $\angle B$ and $\angle C$ will be

- (a) $\angle B = \angle C = 60^\circ$ (b) $\angle B = \angle C = 30^\circ$ (c) $\angle B = \angle C = 45^\circ$ (d) $\angle B = \angle C = 50^\circ$

Ans: (c) $\angle B = \angle C = 45^\circ$

97. The measure of each angle of an equilateral triangle is:

- (a) 60° (b) 30° (c) 45° (d) 40°

Ans: (a) 60°

98. If the vertical angle of a isosceles triangle is 40° then measure of other two angles will be (a) $60^\circ, 60^\circ$ (b) $70^\circ, 70^\circ$ (c) $50^\circ, 50^\circ$ (d) $75^\circ, 75^\circ$

Ans: (b) $70^\circ, 70^\circ$

99. If $\angle A$, $\angle B$ and $\angle C$ of $\triangle ABC$ are equal then triangle is:

- (a) Equilateral (b) Isosceles (c) Scalene (d) None of these

Ans: (a) Equilateral

100. Two line segments are congruent if they have

- (a) Same name (b) Equal measures (c) Unequal measures (d) None of these

Ans: (b) Equal measures

